

Assignment 3 Solutions

3.3 Not possible A has 4 columns and v has only 2 entries

$$3.4 \begin{bmatrix} -6 & -9 & -5 & -3 \end{bmatrix} \begin{bmatrix} 6 \\ -10 \\ 0 \\ -9 \end{bmatrix} = 6(-6) + (-10)(-9) + (-5)(0) + (-9)(-3) \\ = -36 + 90 + 0 + 27 \\ = 81 \quad (\text{or } [81])$$

$$3.5 \begin{bmatrix} -3 & 6 & -18 \\ -1 & 5 & -24 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 6 \\ -1 & 5 & -24 \end{bmatrix} \xrightarrow{\substack{+1 \\ -2}} \begin{bmatrix} 1 & -2 & 6 \\ 0 & 3 & -18 \end{bmatrix} \\ \xrightarrow{\substack{+2 \\ -12}} \begin{bmatrix} 1 & -2 & 6 \\ 0 & 1 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -6 \\ 0 & 1 & -6 \end{bmatrix}$$

$$x_1 = 6$$

$$x_2 = 6$$

$$3.6 \begin{bmatrix} 1 & -4 & 0 & 5 \\ -3 & 12 & -3 & -3 \\ -1 & 4 & -2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 0 & 5 \\ -1 & 4 & -1 & -1 \\ -1 & 4 & -2 & 3 \end{bmatrix} \xrightarrow{\substack{+1 \\ +1}} \begin{bmatrix} 1 & -4 & 0 & 5 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & -2 & 8 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -4 & 0 & 5 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 5 + 4x_2$$

x_2 is free

$$x_3 = -4$$

$$3.11 \quad \begin{bmatrix} 4 & -12 \\ -3 & 9 \end{bmatrix} \sim \begin{bmatrix} 12 & -36 \\ -12 & 36 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 \\ 0 & 0 \end{bmatrix}$$

No, 2 rows but 1 pivot

$$3.12 \quad \begin{bmatrix} 1 & 5 & -1 \\ 2 & -4 & 5 \\ 4 & 0 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & -1 \\ 0 & -14 & 7 \\ 0 & -20 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & -1 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

No, 3 rows but 2 pivots

$$4.2 \quad \begin{bmatrix} 1 & -1 & -3 \\ -1 & 4 & 9 \\ 2 & -3 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -3 \\ 0 & 3 & 6 \\ 0 & -1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = x_3 \quad x = 1$$

$$x_2 = -2x_3 \quad x_2 = -2$$

$$x_3 \text{ is free} \quad x_3 = 1$$

(Lin. Dep.)

$$\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 4 \\ -3 \end{bmatrix} + \begin{bmatrix} -3 \\ 9 \\ -8 \end{bmatrix} = \begin{bmatrix} 1 + 2 - 3 \\ -1 - 8 + 9 \\ 2 + 6 - 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

4.4 Lin. Dep. because there are 4 vectors in $\mathbb{R}^3$

$$\begin{bmatrix} 1 & -1 & -3 & -2 \\ 0 & 2 & 2 & -2 \\ 1 & 0 & 4 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -3 & -2 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -3 & -2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 6 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -3 & -2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$x_1 = x_4$$

$$x_2 = 2x_4$$

$$x_3 = -x_4$$

x_4 is free

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} -3 \\ 2 \\ 4 \end{bmatrix} + \begin{bmatrix} -2 \\ -2 \\ 3 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & -2 & +3 & -2 \\ 0 & +4 & -2 & -2 \\ 1 & -4 & +3 & \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

4.6

$$\begin{bmatrix} 2 & -3 & 3 & 8 \\ 0 & -2 & -2 & 7 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -3 & 3 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -3 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & 0 & 6 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$x_1 = -3x_3$$

$$x_2 = -x_3$$

x_3 is free

$$x_4 = 0$$

$$3\vec{v}_1 + \vec{v}_2 - \vec{v}_3 = \vec{0}$$

3.16 True # pivots $\leq \min \{m, n\} < m$
fewer pivots than rows

3.19 False $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

3.20 True $A \vec{v}_i = \vec{b}$

$$[\vec{a}_1, \dots, \vec{a}_n] \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \vec{b} \Rightarrow$$
$$v_1 \vec{a}_1 + \dots + v_n \vec{a}_n = \vec{b}$$

$$[\vec{a}_1, \dots, \vec{a}_n, \vec{b}_1, \dots, \vec{b}_n] \begin{bmatrix} v_1 \\ \vdots \\ v_n \\ 0 \\ \vdots \\ 0 \end{bmatrix} = v_1 \vec{a}_1 + \dots + v_n \vec{a}_n = \vec{b}$$

3.23 False $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$

4.7 False $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$

4.9 False $\vec{v}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\vec{v}_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

4.12 True Either $\vec{v}_1$ or $\vec{v}_2$ is nontrivial

4.14 False $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ $\vec{v}_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

4.15 True $\alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \alpha_3 \vec{v}_3 = \vec{0} \Rightarrow$

$$\begin{aligned} & \alpha_1 (\vec{v}_1 + \vec{v}_2) + \frac{\alpha_2 - \alpha_1 - \alpha_3}{2} (\vec{v}_1 + \vec{v}_2) + \alpha_3 (\vec{v}_3 + \vec{v}_1) = \\ & \alpha_1 \vec{v}_1 + \alpha_1 \vec{v}_2 + (\alpha_2 - \alpha_1 - \alpha_3) \vec{v}_2 + \alpha_3 \vec{v}_3 + \alpha_3 \vec{v}_1 = \\ & \alpha_1 \vec{v}_1 + \cancel{\alpha_1 \vec{v}_2} + \alpha_2 \vec{v}_2 - \cancel{\alpha_1 \vec{v}_2} - \cancel{\alpha_3 \vec{v}_2} + \alpha_3 \vec{v}_3 + \cancel{\alpha_3 \vec{v}_1} = \\ & \alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \alpha_3 \vec{v}_3 = \vec{0} \end{aligned}$$

3.27 (a)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

(c)

$$\begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & \\ \frac{1}{2} & \frac{1}{2} & & \\ 1 & & & \end{bmatrix}$$

3.30 (a) B is in echelon form, 1 pivot per row

$$(b) \left[\begin{array}{cccccc|c} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right] \sim \left[\begin{array}{cccccc|c} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right] \sim \dots$$

$$\begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

$$4.16 (a) \sim \left[\begin{array}{cccc} 1 & 1 & 3 & -2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cccc} 1 & 1 & 0 & -5 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cccc} 1 & 0 & 0 & -9 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\vec{v}_4 = -9\vec{v}_1 + 4\vec{v}_2 + \vec{v}_3$$

(b) Yes, Lin. Ind.

$\{\vec{v}_2, \vec{v}_3\}$ is L.I. because not colinear

$$x\vec{v}_2 + y\vec{v}_3 = \vec{v}_4 \Rightarrow y = 1 \quad (\text{by row 3})$$

$$\Rightarrow x = 4 \quad (\text{by row 2})$$

$$\Rightarrow 7 = -2 \quad (\text{by row 1})$$

contradiction

$$\text{so } \vec{v}_4 \notin \text{span} \{ \vec{v}_2, \vec{v}_3 \} \Rightarrow$$

$$\{ \vec{v}_2, \vec{v}_3, \vec{v}_4 \} \text{ is L.I.}$$

4.17

$$\begin{aligned} \begin{bmatrix} 1 & 1 & -2 \\ 9 & 4 & h \\ -3 & 3 & -6 \\ -1 & 1 & -2 \\ +1 & +1 & -2 \end{bmatrix} &\sim \begin{bmatrix} 1 & 1 & -2 \\ 0 & -5 & h+18 \\ 0 & 2 & -4 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & -5 & h+18 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & h+8 \end{bmatrix} \end{aligned}$$

$$h = -8$$

4.23

$$x(\vec{v}_1 + \vec{v}_3) + y(-2\vec{v}_1 + \vec{v}_2 + \vec{v}_3) + z(-3\vec{v}_1 - \vec{v}_2 - 6\vec{v}_3) = \vec{0}$$

$$(x - 2y - 3z)\vec{v}_1 + (y - z)\vec{v}_2 + (x + y - 6z)\vec{v}_3 = \vec{0}$$

L.I. $\Rightarrow$

$$x - 2y - 3z = 0$$

$$y - z = 0$$

$$x + y - 6z = 0$$

$$\begin{bmatrix} 1 & -2 & -3 \\ 0 & 1 & -1 \\ 1 & 1 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -3 \\ 0 & 1 & -1 \\ 0 & 3 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x = 5z$$

$$y = z$$

z is free

$$5\vec{u}_1 + \vec{u}_2 + \vec{u}_3 = \vec{0}$$